

Data Mining Analysis of Qualifying Performance as a Predictor of Race Outcomes in Formula One

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Abstract

This study corroborates and builds upon the previous work on the predictive model of qualifying performance and race results in F1 over the past 22 seasons (2006 to 2023) with 400 races and more than 7,800 driver-weekend observations. The analysis, conducted with three statistical approaches (Contingency Coefficient [C], Spearman's Rank Correlations [ρ] and Ordinal Logistic Regression [OLR]), shows that the single strongest predictor is the 'qualifying position'. The C values consistently were greater than 0.81 for all of the seasons examined and Spearman's ρ was consistently greater than 0.70 with the OLR model having a statistically significant coefficient ($p < 0.001$) and a McFadden pseudo- R^2 of 0.34. In addition, a novel Team Reliability vs. Pace Quadrant Analysis is suggested to overcome a major limitation of the original study, namely that it did not consider DNF (Did Not Finish) events in the qualifying-race correlation that would cast doubt on the original analysis. This team performance analysis methodology reveals actionable strategic insights beyond what can be gained from raw correlation analysis - the methodology is split into pace (average qualifying position) and reliability (DNF rate). Qualifying-race results corroborate the strength of the qualifying-race relationship throughout all of the sport's eras.

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1.0 Introduction

1.1 Background of the Study

Formula One (also known as F1) is among the most data-driven sports in the world (Weissbock & Mills, 2025), where it consists of a 3 day ‘race weekend’ generating countless data on performance indicators from the structured format of: 2 practice sessions on Fridays, qualifying sessions on Saturdays and the Grand Prix itself, which is the final race on Sundays. Each of these stages contribute to the overall race result of a driver. The reliance of this sport on cutting-edge technologies makes data highly significant for the competitiveness of the team. In order to achieve success, teams analyse aerodynamic efficiency of the car, pit-stop strategy, tyre degradation, weather and altitude during the race as well as competitor data so that they can optimise race strategies, car setups and in-race decisions (Fatima & Johrendt, 2023).

In particular, the qualifying session is considered a critical determinant of success as it determines and sets the grid position for a driver which dictates early race strategy. However, despite its importance, there is an ongoing debate on whether it’s accurate that these qualifying results predict the final race positions as other factors like; race incidents and pit-stop efficiency can alter these expected outcomes.

This study is based on the case study “*Evaluating the Predictive Power of Qualifying Performance in Formula One Grand Prix*” (Joshua Weissbock & Dr. Mills, 2025). The research analyses nearly two decades of race data (2006-2023) in order to assess the predictive value of qualifying results using data mining techniques such as contingency coefficients and Ordinal Logistic Regression.

1.2 Aim of the Case Study

The main aim of this study is to explore the predictive relationship between qualifying performance and race outcomes in F1. Through statistical analysis and data mining, the case study evaluates whether qualifying results can be used as a reliable predictor of finishing position.

Furthermore, this research further compares qualifying data with other factors such as starting grid positions to identify which variable holds the highest predictive strength for race performance.

1.3 Problem Statement

In Formula One, teams invest heavily in qualifying performance under the assumption that a strong grid position equals a higher chance of race success. However, this is not the sole

factor to success when other factors can alter race dynamics and diminish the advantage of a strong grid position. Therefore, the core problem addressed in this study is:

“To what extent does qualifying performance predict final race outcomes in F1 and how consistent is this relationship across different seasons and circuits?”

1.4 Objectives

The objectives of the study is:

- To preprocess and explore Formula One race and qualifying data from 2006-2023 for accurate analysis.
- To validate the case study’s core findings through implementation.
- To propose an improved predictive framework for race performance modeling.
- To evolve understanding of car performance over a full weekend to maximise performance in the Grand Prix race.

2.0 Literature Review

As the rapid advancement of data mining in sports analytics continues, it opens doors for researchers to uncover patterns in large, complex datasets especially in Formula one where data mining plays a crucial factor in predicting performance, identifying influential factors and optimising strategies.

In order to understand the background and significance of this study, previous research on race prediction and qualifying analysis is reviewed. This following section examines ten studies that showcase the use of data mining or machine learning for race prediction.

Historical and earlier studies within F1 have established that qualifying performance strongly correlates with race results, but the degree of this correlation changes depending on the regulations that season as well as the circuit characteristics. Methods such as statistical analysis and machine learning models have both been used in order to make the evaluation of the relationship.

2.1 Related Research

Early research into F1 race prediction relied on classical statistical methods Muehlbauer (2010) gave the first quantitative confirmation that the starting grid position is strongly correlated with race outcome ($C = 0.63$, $p < 0.001$), which was refined by subsequent studies. This was expanded by Phillips (2014), who showed that statistical modelling could factor out driver skill from car performance over 60 years of driving history, without the necessity of machine learning. Patil and Jain (2022) used PCA to reduce 21 performance variables to 4 principal components that accounted for about 70% of the variance of the season points. Overall, these studies confirm that pre-race variables are important for predicting race outcomes, and dimensionality reduction is feasible in F1 data.

The use of machine learning greatly enhanced prediction accuracy. Rao (2024) showed that Random Forest models can predict podium finishes with 87% accuracy from knowledge of grid position, constructor and meteorological data. Urdhwareshe (2025) reported better performance by TabNet than Random Forest and Gradient Boosting in 4,637 race observations, which reinforces the fact that pre-race variables are good predictors. Stoppels (2017) has demonstrated earlier that Artificial Neural Networks obtained a moderate level of accuracy, qualifying position, and team strength being crucial inputs. The focus of the present study is in line with the findings of all machine learning studies, which is the qualifying position among the highest predictor variables, irrespective of the algorithm used.

Despite the wide variety of these methods, there is a clear gap – very little study explicitly considers the difference between the qualifying speed and the ability of a car to complete the

race. Fatima and Johrendt (2023) and Todd, García and Schaefer (2025) use variables related to reliability (like tyre degradation, pit stop data) but do not consider DNF rate as an additional dimension of reliability. This study tackles that issue directly by proposing the Quadrant Analysis that breaks team performance into two measurable dimensions: pace and reliability.

2.2 Summary of All Research

The reviewed studies collectively demonstrate the evolution of data mining and machine learning applications in Formula One analytics. Researchers have employed a range of methods to predict race outcomes, driver performance and race strategies. These methods go from the traditional statistical models to advanced neural networks. This section summarises the key techniques, focus areas and main findings of each study so that the contributions of different analytical approaches are highlighted to understand and improve performance in Formula One racing.

Author(s)	Technique Used	Focus Area	Key Outcome
Fatima & Johnrendt (2023)	Embedded Deep Neural Network (EDNN)	Predicting race-winning strategies with tyre degradation, weather and pit stop data	Including environmental and strategic factors improved prediction accuracy compared to traditional models
Rao (2024)	Random Forest, XGBoost, Logistic Regression	Predicting race results from grid constructor, circuit and weather data	Random Forest achieved 87% accuracy; qualifying data and team performance strongly influence race outcomes
Surendran (2024)	LSTM and GRU Deep Learning Models	Predicting driver and team rankings across seasons	Qualifying and circuit characteristics are consistent predictors of race success
Urdhwareshe (2024)	TabNet, Random Forest, Gradient Boosting	Predicting race results and championship standings	TabNet performed the best. Found that pre-race variables are strong indicators of race success
Todd, Garcia & Schaefer (2025)	XGBoost with Explainable AI	Predicting tyre energy and performance using telemetry and environmental data	Tyre compound, track temperature and stint length were key predictors. The use of telemetry improves strategy decisions
Choo (2020)	In-race Modelling Techniques	Predicting F1 and NASCAR performance	Sector-level modelling can forecast race

		at lap and sector levels	outcomes and in race adjustments effectively.
Muehlbauer (2010)	Contingency Coefficient Correlation Analysis	Relationship between starting and finishing positions	Found strong positive correlation ($C = 0.63$, $p < 0.001$). Pole position drivers had around 60% win rate
Phillips (2014)	Statistical Modelling	Adjusting for team and competition effects to compare driver performance	Driver skill and consistency play major roles in success beyond car performance
Stoppels (2017)	Artificial Neural Network (ANN)	Predicting race results through the use of historical motorsport data	ANN achieved moderate accuracy. Qualifying and team strength are the key influencing factors of race results.
Patil & Jain (2022)	Correlation and Principal Component Analysis (PCA)	Identifying variables influencing total season points for a driver	Four principal components explained ~70% variance. The top factors (influencers) were completed races and car performance.

Table 1.0: Summary table of references used.

3.0 Dataset

This section details the data acquisition, cleaning, preprocessing as well as the feature election methods employed in the case study. All of the procedures applied ensures consistency, accuracy and suitability for the analysis of the study, which evaluates how qualifying session metrics influence final race outcomes.

3.1 Data Acquisition and Collection

The data for this study was compiled from multiple official Formula One race data spanning the 2006 to 2023 seasons and a total of results from 400 races. The raw data consists of results from practice sessions, qualifying sessions and the final race, as well as the starting grid positions which represented each driver's complete race weekend. This data was then compiled into four main CSV files creating a dataset that encompasses over 7,800 driver-weekend observations.

Dataset	Description
fl_results_practices.csv	Contains results from the practice sessions
fl_results_qualifications.csv	Contains the results of the qualifying sessions
fl_results_races.csv	Contains the final race outcomes and results
fl_starting_grid.csv	Contains the starting grid positions for each race
copy_final_fl_datav2.csv	Final dataset - made from the integration of the 4 datasets

Table 2.0: Table of all datasets used

3.2 Data Preprocessing and Cleaning

The raw data from the four CSV files required significant preprocessing as well as cleaning in order to ensure consistency, remove redundancies and create a unified, analysis-ready dataset to develop the prediction model. The final dataset in this case study was called copy_final_fl_datav2.csv.

3.2.1 Data Filtering and Exclusions

In this study, the initial step focused on reducing noise and ensuring data comparability in the dataset by filtering out non-standard events and incomplete records. Two main filters and exclusions were made to maintain consistent data structure which was the exclusion of atypical race weekends and exclusion of incomplete driver entries.

For the exclusion of atypical race weekends, specific Grand Prix events were entirely removed if they were cancelled or incomplete. For example, the 2023 Emilia Romagna Grand Prix which was cancelled due to flooding. Furthermore, sprint race weekends, which were newly introduced in 2021 where the weekend features a different format with a shorter “sprint race” on Saturdays, were excluded as the altered structure and reduced practice time changes the teams’ strategic approach.

To ensure that the analysis focuses on a consistent driver-car pairing throughout the weekend, individual records were dropped if a driver did not participate in all three sessions (practice, qualifying, race) or if a session was significantly impacted by external factors like weather which prevent meaningful participation. As a result, this addresses scenarios like session-specific issues and mid-weekend driver swaps that would compromise the integrity of that record.

3.2.2 Feature Renaming

To improve clarity and prevent ambiguity across the dataset and when they merge, columns were renamed. The raw datasets all contained a generic Pos column for the finishing position which was a critical issue as distinguishing it from the qualifying position after merging would be impossible. Which is why each Pos column was renamed to be specific to its session before merging. For example, Pos in the qualifying data became Post_q and Pos in the race data became Pos_race. This ensures that every feature in the final dataset has a unique and descriptive name.

3.2.3 Feature Selection

Feature selection is the process of identifying the most relevant variables that play an important role in determining the outcome variable. The selection process of this case study was driven by the goal of predicting race outcomes thus the chosen variables capture a driver’s performance throughout a race weekend and are grouped into three main categories; predictor variables (eg; qualifying performance), target variables (eg; race position) and identifier variables (eg; season, GP, driver). The key features for each row in the dataset for each session is as below:

Feature Name	Description
Season	The year of the race
GP	Grand Prix, the name of the Grand Prix
gp_code	Unique identifier code F1 gives to that F1 weekend
No	Driver’s number

Car	The team or manufacturer the driver represented
Gap (For each session)	Gaps during each practice session compared to the session leader
Laps (For each session)	The number of laps completed by the driver during the session
Pos (For each session)	The final ranking position determined during that session
Time (For each session)	The best lap time the car achieved in that session
Qualifying Time	The best time the car achieved during each stage of the qualifying session
Race Start Position	Adjusted grid position accounting for penalties or disqualifications
Race Finish Position	Final result of the driver in the race
Status	Whether the driver finished the race (and how far behind the leader) or did not finish (DNF) and the reason IF applicable
Points	Points awarded to the driver for their finishing position

Table 3.0: Table of feature names

3.2.4 Data Transformation (Pivoting) and Integration

The raw practice session data from `f1_results_practices.csv` was in a “long” format where each driver had multiple rows corresponding to each practice session – Practice 1 (P1), Practice 2 (P2) and Practice 3 (P3). In this study, the data was pivoted into a “wider” format to create a single, comprehensive record for each driver per Grand Prix. This transformation converts the session-specific rows into columns (eg; `Pos_P1`, `Time_P1`, `Pos_p2`, `Time_p2`, etc.) which, as a result, consolidates all driver’s practice metrics into a single row. Finally, the cleaned data from the four separate files were merged into a single, unified DataFrame where two dataframes merge one by one through a left join.

3.2.5 Handling Missing Values

Missing values were handled with the following strategies: rows with missing values in the `Pos_race` column were dropped entirely, as the target variable must be present for any record to contribute to the analysis. For text-based fields such as lap times (Q1, Q2, Q3), missing values were filled with a "No Time" placeholder. All other positional columns retain NaN values, which are excluded pairwise during statistical analysis to avoid corrupting correlation results.

4.0 Proposed Methods

To analyse the relationship between various Formula One sessions and the final race outcome, this study adopts the statistical methods employed in the selected case study. The main objective is to quantify the predictive strength of practice sessions, qualifying and starting grid position on a driver's final race position. The methodologies used in this study involve multi-layered analysis to generate results and identify trends over time. The 3 methodologies are Contingency Coefficients (C), Ordinal Logistic Regression (OLR) and Spearman's rho (rank correlation).

4.1 Contingency Coefficients (C)

Contingency Coefficient (C) is a statistical measure used to determine the strength of the association between two nominal/categorical variables – specifically, the relationship between the starting grid position and the finishing position. C is a measure of the degree of association between two variables, with a value ranging from 0 to 1; with 0 being an indication that the two nominal variables are independent of each other (Muehlbauer, 2010). This method constructs contingency tables to analyse the frequencies of the outcomes between pairs of sessions. For example, the table would compare qualifying and race positions to see how many drivers who qualified in the “1st place” category also finished the race in the “1st place” category, the “2nd place” category and so on. This method allowed the study to quantitatively prove if the outcome of one session was statistically dependent on another. A high coefficient value between Qualifying and Race would state that a strong and predictable pattern exists and serve as the foundational evidence for the paper's main argument. The conclusion drawn from this method is that the category a driver finishes in for the race is most strongly dependent on the category they achieved in qualifying thus providing core evidence that underpins the entire study. In this study, the Contingency Coefficient between qualifying position and race position was calculated as $C = 0.7731$ across the full 2006–2023 dataset, with seasonal values ranging from 0.8163 (2023) to 0.8684 (2007).

4.2 Spearman's Rho (Rank Correlation)

The case study calculated the Spearman's rho for all session positions to rank-order correlations and create its ‘coefficient table,’ which will be visualised as a heatmap later in this study. While C captures the strength of nominal associations, Spearman's Rho, on the other hand, assessed nonparametric and monotonic relationships between ranked variables. In this case, it's used for qualifying/starting/finish positions. This metric is used in addition with C which complements each other as Spearman can take account rank ordering patterns. The analysis revealed that Qualifying Position (Pos_q) consistently has the highest correlation

coefficient with the Final Race Position (Pos_race) which is a key finding as it implies that raw qualifying performance is a more powerful indicator of a driver's potential on race day than their final grid spot. The overall Spearman's ρ between Pos_q and Pos_race across the full dataset was 0.7750 ($p < 0.001$, $n = 6,005$).

4.3 Ordinal Logistic Regression (OLR)

The final step of the analysis in the case study employed the OLR model to showcase the relationship between the predictor variables (practice session position and qualification position) and the target variable (race finishing position categorised in ordinal groups eg; 1st, 2nd,...). OLR is appropriate, especially in the context of this study, as finishing position is an ordered category – this allows quantifying the relationship between practice and qualifying sessions with the final race performance. The results from this study validates its initial findings but provides a more rigorous conclusion that reinforces the “dominant role” of qualifying in determining race outcomes. The OLR model confirmed the qualifying position as a statistically significant ordinal predictor ($\beta = -0.3629$, $p < 0.001$, McFadden $R^2 = 0.3407$).

5.0 Implementation of Dataset

Following the established methodologies from above, this section transitions from the theory application to the practical by leveraging the fully preprocessed dataset (copy_final_f1_datav2.csv). The statistical techniques discussed will be implemented to test and visualise the core hypothesis of the case study and to validate the case study's findings by replicating its key analysis.

5.1 Contingency Table and Heatmap Analysis

The Contingency Table depicts the relationship between qualifying positions (Pos_q) and race positions for the 2010-2023 season through the use of a heatmap. The heatmap provides a clear two-dimensional view of the outcomes with each cell representing a specific qualifying-to-race-finish combination as well as colour intensity to show the frequency (or count) of an event occurring. This method reveals the most common outcomes and the strong central tendency for drivers to finish near where they start.

To generate this heatmap from the case study on the final f1 dataset, we first filter the dataset to include only the seasons from 2010 to 2023. Then, the columns for Pos_q and Pos_race are cleaned by removing any non-finish classification (ensures all values are numeric). Then, a contingency table is made that counts the occurrences of each qualifying-to-race position pair which is then plotted as a heatmap. Below, in Figure 5.1.1, is the ready visualised heatmap and some key parts of the code.

The heatmap provides a detailed validation of the case study's primary conclusion. One of the key interpretations is the clear and strong diagonal trend. The blue line running diagonally from the top-left to the bottom-right has a clear concentrated blue for Qualifying 1st and Finishing 1st, confirming graphically the extremely high correlation between qualifying and finishing positions. The most intense colours are also clustered tightly around this main diagonal which demonstrates that small deviations for a driver is normal (losing or gaining a few positions). In conclusion, we can conclude that qualifying provides an 'unbiased' measure of driver and car capability as this capability translates directly into the final race result (Weissbock & Mills, 2025).

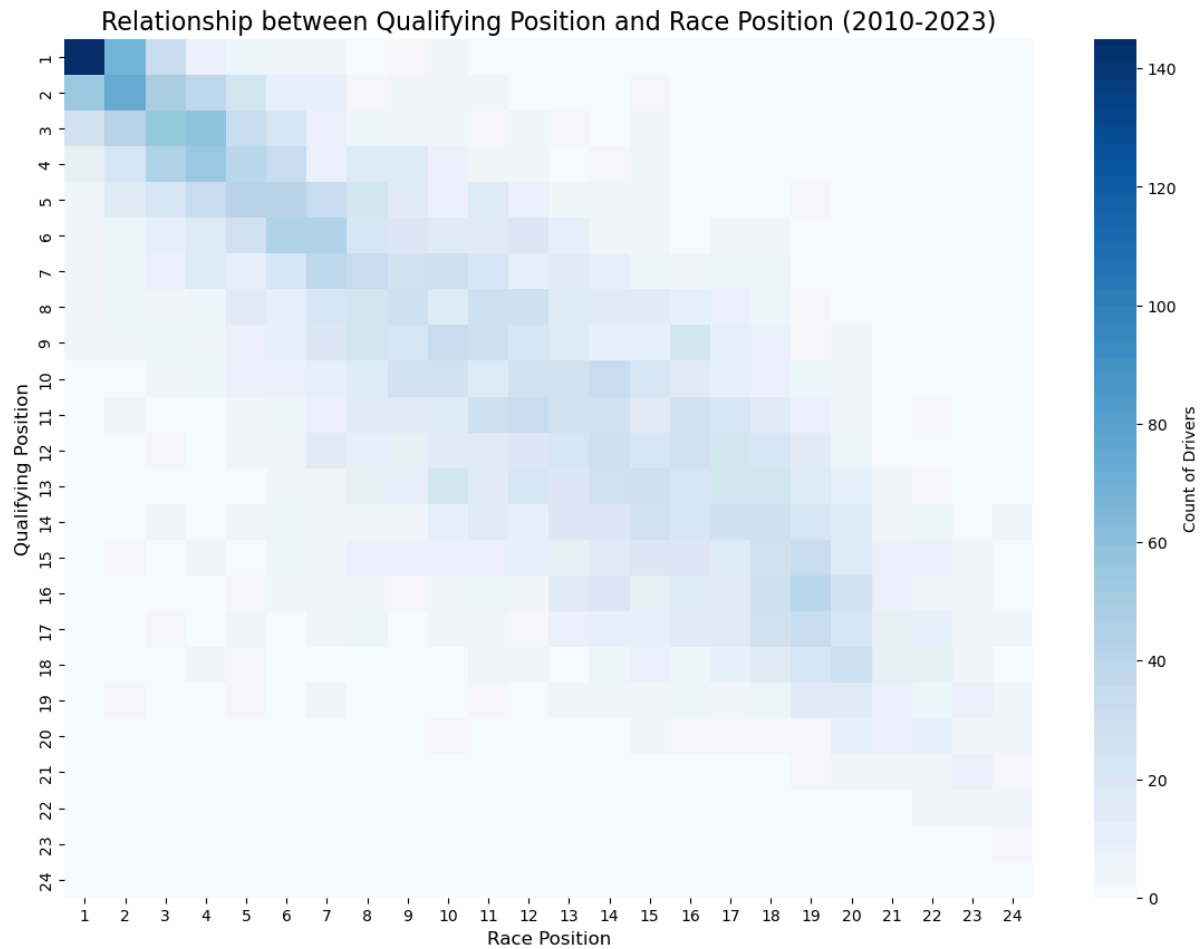


Figure 5.1.1: Heatmap

5.2 Season by Season Analysis

This analysis provides the most granular view of the relationship between qualifying and race results by breaking down the correlation on a year-by-year basis. The case study outlined that the purpose of the analysis is to examine the consistency of qualifying's predictive power over time using both the Spearman's Rank Correlation (ρ) as well as the Contingency Coefficient (C). By plotting both coefficients for each individual season, we are able to identify specific years that were more or less predictable and understand factors that affect the race outcome such as team dominance.

To create the visualisation like the case study, data cleaning is done to handle non-numeric entries. Then for each season, we calculate the Spearman's Rho (directly from the Pos_q and the Pos_race columns) as well as the Contingency Coefficient. Finally, both sets of the results are plotted as line graphs. Below in Figure 5.2.1 is the line graph, where we can interpret the consistent strong correlations across all seasons with C values exceeding 0.81 and ρ over 0.70. Overall, the line graph proves that the strong link between qualifying and race results is not an average, rather it has been true every single year. All seasonal correlations were

statistically significant at $p < 0.001$, and the overall Spearman's ρ across the full 2006–2023 dataset was 0.7750 ($p < 0.001$, $n = 6,005$). As mentioned in the case study, the connection is even stronger in years when one team dominates (like Mercedes in 2015). The strongest correlation was observed in 2011 ($\rho = 0.8510$), a season of significant Red Bull dominance, while 2023 showed the weakest ($\rho = 0.7060$), reflecting the more competitive multi-team field of recent seasons. This demonstrates that qualifying is a reliable predictor across all of F1's different technical eras.

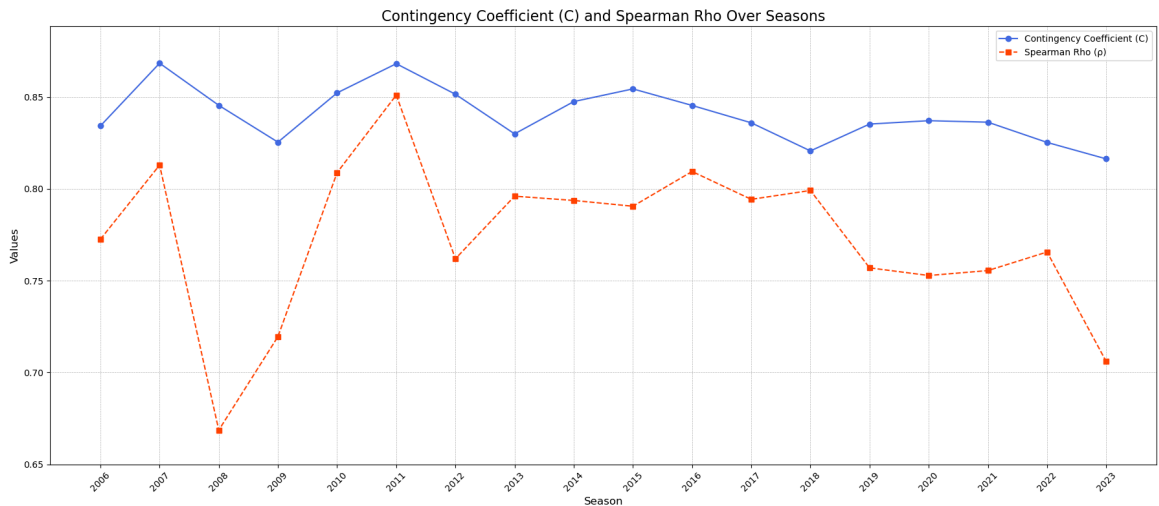


Figure 5.2.1: Season by Season Analysis Visualisation

--- Seasonal Correlation Results ---					
Season	Spearman ρ	p-value (ρ)	Significant	Contingency C	
2006	0.7726	0.0	Yes	0.8343	
2007	0.8129	0.0	Yes	0.8684	
2008	0.6686	0.0	Yes	0.8454	
2009	0.7195	0.0	Yes	0.8254	
2010	0.8088	0.0	Yes	0.8522	
2011	0.8510	0.0	Yes	0.8681	
2012	0.7619	0.0	Yes	0.8515	
2013	0.7960	0.0	Yes	0.8299	
2014	0.7937	0.0	Yes	0.8475	
2015	0.7905	0.0	Yes	0.8544	
2016	0.8094	0.0	Yes	0.8454	
2017	0.7943	0.0	Yes	0.8359	
2018	0.7991	0.0	Yes	0.8206	
2019	0.7570	0.0	Yes	0.8352	
2020	0.7528	0.0	Yes	0.8371	
2021	0.7555	0.0	Yes	0.8363	
2022	0.7656	0.0	Yes	0.8253	
2023	0.7060	0.0	Yes	0.8163	

Figure 5.2.2: Seasonal Spearman ρ and Contingency C values (2006–2023). All p-values = 0.000.

5.3 Multi-Variable Session Correlation Analysis

A Spearman rank correlation matrix was generated for all five positional variables: Practice 1, Practice 2, Practice 3, Qualifying, and Race Start Position, to determine what session had the highest correlation with the final race classification. A correlation matrix (Figure 5.3.1) is created, and this shows that the qualifying position (Pos_q) has the highest correlation with the final race position with the value being 0.752, slightly higher than the Race Start Position ($\rho = 0.742$). There were also a decreasing number of correlations between the practice sessions (Practice 3: $\rho = 0.521$, Practice 2: $\rho = 0.588$, and Practice 1: $\rho = 0.419$). The difference between the qualifying (0.752) and best practice session (P2 at 0.588) was significant, and reflected the convergence and refinement of car and driver performance over the weekend, where qualifying was the best indicator. The fact that Pos_q and Race_Start_pos have almost identical correlation (0.752 versus 0.742) is also a sign of the high co-linearity of these two variables: in most cases it is the same persons who qualify for the race and who will take part in this race.

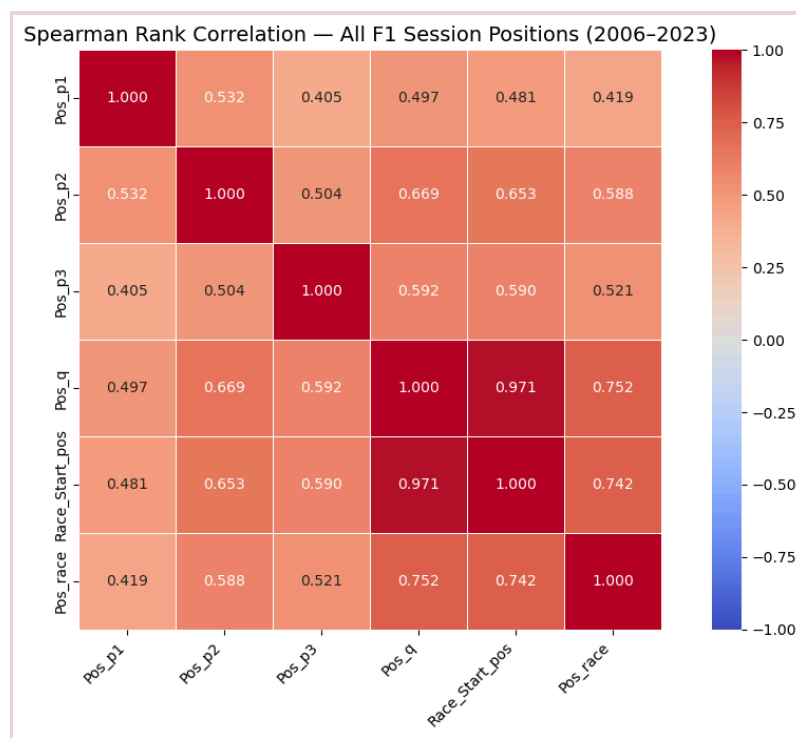


Figure 5.3.1: Spearman correlation matrix across 5 positional session variables against final race position

5.4 Ordinal Logistic Regression Results

An Ordinal Logistic Regression model was fitted to the entire set of 2006–2023 data to give a more stringent validation of the predictive power of qualifying. The top three places were considered as "Podium", the next six as "Points" and the remaining as "Outside Points". Position (Pos_q) was the only predictor variable.

The model produced a statistically significant coefficient for qualifying position ($\beta = -0.3629$, $SE = 0.007$, $z = -49.91$, $p < 0.001$). The negative sign is to be expected: in this model, a worse (lower) qualifying position increases the log-odds of achieving a higher outcome category in the race. The log-odds of improving the finishing category drop by 0.3629 for each position drop in qualifying (e.g. P1 to P2). The McFadden pseudo- R^2 is very high at 0.3407, which represents a good fit for the model (values > 0.20 are excellent fit in an ordinal logistic regression). Statistically significant, the model is also practically meaningful with 6,005 observations in the dataset from 2006 to 2023, accounting for about 34% of the variance in race outcome category.

Parameter	Coefficient (β)	Std. Error	z-score	p-value	95% CI
Qualifying Position (Pos_q)	-0.3629	0.007	-49.908	< 0.001	[-0.377, -0.349]
Threshold: Outside Points / Points	-4.4572	0.091	-49.014	< 0.001	[-4.635, -4.279]
Threshold: Points / Podium	1.2114	0.020	61.302	< 0.001	[1.173, 1.250]
McFadden R^2	0.3407	–	–	–	–
Observations	6,005	–	–	–	–

Table 4.0: Results of the OLR.

In Table 4.0, the two threshold parameters define the cut-points between the three ordinal categories in the model's probability space. The threshold between "Outside Points" and "Points" ($\beta = -4.457$) and between "Points" and "Podium" ($\beta = 1.211$) are both statistically significant ($p < 0.001$), confirming that the three race outcome categories are meaningfully distinct and that the ordinal structure of the model is appropriate.

6.0 Proposing a New or Improved Output

The case study provides a powerful and robust validation that qualifying performance is the primary predictor of final race outcomes. However, there's a key limitation of the study: it proves that qualifying pace predicts finishing position but doesn't take account of the separation between two main components of a race outcome: the car's speed (pace) and its ability to actually finish the race (reliability). There is an argument where a team can have the fastest car on the grid but still get poor results if it constantly breaks down. Therefore to address this trade-off, I propose an improved analysis for a better output which provides a more complete picture of team performance.

6.1 Team Reliability vs. Pace Quadrant Analysis

This analysis proposes a new output that enriches the case study's findings by decomposing race performance into two core components: raw speed and mechanical reliability. While the case study establishes a strong link between qualifying pace and the final race rank, it doesn't account for instances where a high-ranked qualifier fails to finish the race (DNF - Did Not Finish) due to technical failure.

The new model calculates two key metrics for each team: the average qualifying position (a measure of a team's pure, one-lap pace) and DNF rate (the percentage of a team's race starts that end in retirement). Plotting these two metrics against each other on a scatter plot (Figure 6.1.1) creates a performance quadrant which categorises teams into four distinct strategic profiles. The scatter plot is separated into quadrant zones: Fast & Reliable (Championship Zone), Slow & Reliable (Points Finishers), Fast but Fragile (High Risk/Reward) and Slow & Unreliable (Development Zone).

The correlation of a driver's qualifying position with his or her retirement from the race will be compromised by that driver's retirement. A driver qualifying in one position and then retiring from the race will be an unclassified position and thus disrupt the expected pattern. This can be seen in the figure 6.1.1. In the 2014-2021 period, Lotus Mercedes' average qualifying race position was the highest, and nearly one in five race starts was a DNF, with a DNF rate of 39.5%. McLaren Honda (31.7% DNF rate) also had the well documented unreliability of their Honda power unit to help them translate their qualifying pace into race results. Naturally, Red Bull Racing TAG Heuer (21.8% DNF rate) is in this quadrant, which is why their championship challenge in this period was more unpredictable than implied by their impressive raw pace. Meanwhile, Mercedes team-mates finished on the top of the grid for the quickest average qualifying time in the period, but also had the second-lowest rate of DNF's for the team (6.6%). This directly explains their seven consecutive Constructors' Championships. Alfa Romeo Racing Ferrari (8.3% DNF rate) and McLaren Mercedes (4.9%

DNF rate) are the two Slow but Reliable profiles — those that would regularly do well to finish a race but simply weren't quite quick enough to take the points. The Fast but Fragile team are making their priority quality and mechanical strength, whereas the Slow but Reliable team are prioritising aerodynamic and power unit development to catch up.

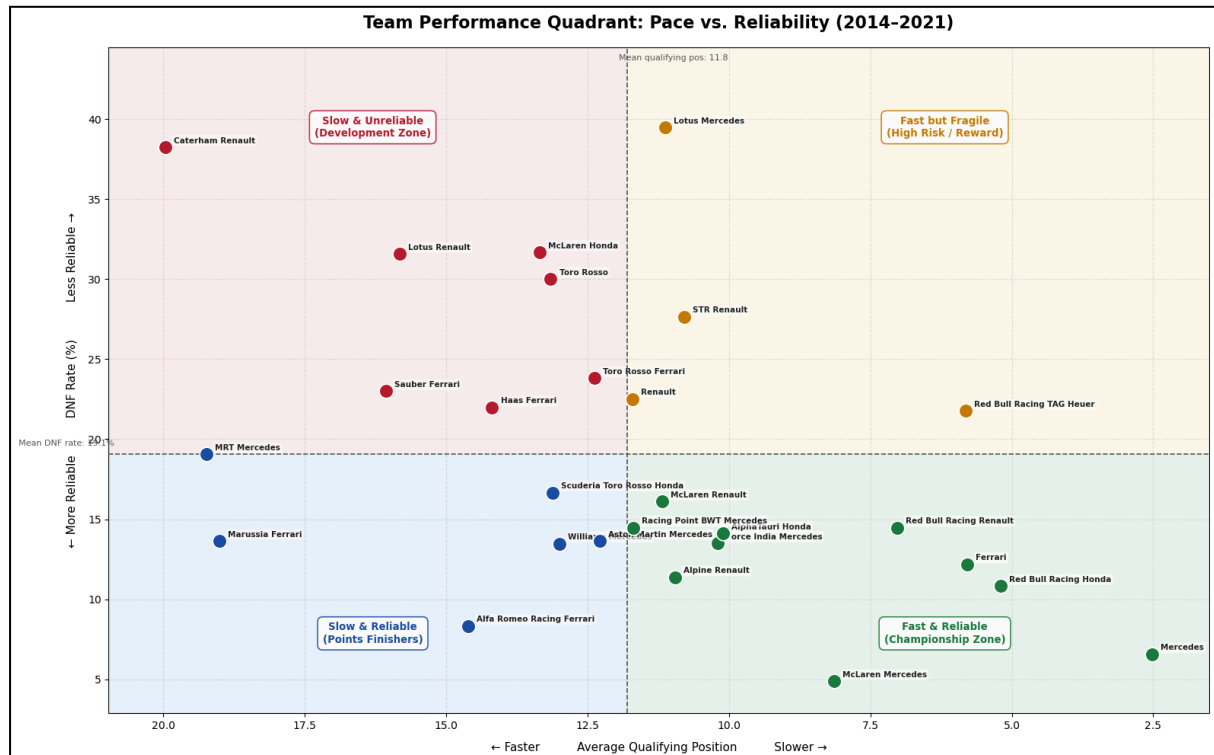


Figure 6.1.1: Quadrant Analysis of Team Performance (2014-2021)

7.0 Conclusion

The objective of this study was to check the predictive power of the relationship between qualifying results and race results in Formula One from 2006 to 2023. The analysis was carried out by applying three complementary statistical methods and included a rigorous preprocessing of more than 7800 driver-weekend observations to test this relationship.

The conclusions are clear. The Spearman rank correlation between qualifying position and race finishing position was $\rho = 0.7750$ ($p < 0.001$, $n = 6,005$) overall and consistently > 0.70 for each of the individual seasons analyzed. The values of contingency coefficient were acceptable (C between 0.8163 and 0.8684) for the various seasons, the average being 0.7731, showing that there was a very strong categorical association between qualification and race outcome. This was further confirmed by the Ordinal Logistic Regression model, which showed a statistically significant qualifying position coefficient ($\beta = -0.3629$, $p < 0.001$) and a McFadden pseudo- R^2 of 0.3407, suggesting that the model accounted for around 34% of the variance in podium finish, points finish and non-points finish.

The multi-variable correlation analysis also showed that, out of all weekend sessions, qualifying is the most predictive data point ($\rho = 0.752$ vs Pos_race), with better predictive value than Race Start Position ($\rho = 0.742$), Practice 2 ($\rho = 0.588$), Practice 3 ($\rho = 0.521$) and Practice 1 ($\rho = 0.419$).

The main drawback of the original case study is the lack of explicit modelling of a confounding variable, the DNF rate, as a factor in the Team Reliability vs. Pace Quadrant Analysis. This is a good way of showing that mechanical reliability is a different aspect of team performance to raw qualifying pace, and that despite being competitive at qualifying, teams that are prone to most often seeing the red flag — most notably McLaren Honda with an average of over 30% DNFs in the 2014–2021 period — cannot convert grid positions to race results at the same rate as teams in the "Championship Zone". However, Mercedes achieved the quickest mean qualifying number and one of the lowest DNF rates in that period, a direct reflection of their dominance during that time of their history.

Qualifying performance remains the single strongest quantifiable predictor of F1 race outcomes available before the race begins. Future work should incorporate safety car frequency, weather conditions, and pit-stop strategy variance to model the residual variance that qualifying alone cannot explain.

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